Day 005 | Unit 01 | Teacher plan

Positive Rational Numbers: Decimals and Division

Estimate and Check Whole-Number Division

Standards: 6.NS.B.2

Learning target: I can divide multi-digit whole numbers and check the quotient.

enVision Topic 1: Use Positive Rational Numbers | Spectrum companion Chapter 1

0–5 minutes / Launch

Read the target. Ask for an estimate or a sketch before calculating. Connect the new idea to the previous lesson.

5–17 / Model

Use Worked Example 1, then Example 2. Ask students to name the next step and explain a choice. Estimate the quotient first. Divide, multiply, subtract, and bring down one digit at a time. Keep each quotient digit in its place; write zero when needed. Check dividend = divisor × quotient + remainder.

17–42 / Practice

Assign questions 1–10 in order. Use 1–3 for foundations, 4–7 for application, and 8–10 for state-format response and reasoning. The example button provides a separate question with a worked method.

42–50 / Discuss and check

Compare representations. Review a selected answer and a plausible error. Use question 10 as an exit discussion; ask students to check units, scale, or substitution.

Support and extension

Offer number lines, fraction strips, grid paper, or a table template as appropriate. Read multi-part directions one part at a time. For extension, ask for another representation or a counterexample.

Companion and state alignment

Use the matching skill in the edition you own. No unverified publisher page numbers are assigned. AASA references below identify related public sample items by standard and format; all course questions are original.

AASA sample item references: 12

Day 005 | Unit 01 | Lesson

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01 / Learn

Estimate the quotient first. Divide, multiply, subtract, and bring down one digit at a time. Keep each quotient digit in its place; write zero when needed. Check dividend = divisor × quotient + remainder.

Worked example 1

Use long division: 175 ÷ 5. Show the algorithm and check.

1. Estimate how many digits the quotient should have. Work from the leftmost digits of the dividend.

2. 17 ÷ 5: write 3, subtract 15, remainder 2

3. bring down the next digit: 25 ÷ 5: write 5, subtract 25, remainder 0.

4. Check: 5 × 35 = 175.

Answer: 35

Worked example 2

Use long division: 5,554 ÷ 35. Show the algorithm and check.

1. Estimate the quotient first.

2. 55 ÷ 35: write 1, subtract 35, remainder 20

3. bring down the next digit: 205 ÷ 35: write 5, subtract 175, remainder 30

4. bring down the next digit: 304 ÷ 35: write 8, subtract 280, remainder 24.

5. Check: 35 × 158 + 24 = 5554.

Answer: 158 R 24

Day 005 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Use long division: 729 ÷ 9. Show the algorithm and check.

02 / Constructed response

Use long division: 252 ÷ 4. Show the algorithm and check.

03 / Constructed response

Use long division: 430 ÷ 5. Show the algorithm and check.

04 / Constructed response

Use long division: 2,299 ÷ 19. Show the algorithm and check.

05 / Constructed response

Use long division: 2,716 ÷ 14. Show the algorithm and check.

06 / Constructed response

Use long division: 5,928 ÷ 38. Show the algorithm and check.

07 / Constructed response

Use long division: 6,701 ÷ 21. Show the algorithm and check.

08 / Model and constructed response

Use long division: 2,368 ÷ 15. Show the algorithm and check.

09 / Model and constructed response

Use long division: 11,312 ÷ 37. Show the algorithm and check.

10 / Constructed response

A club has 4822 cards. Each full packet holds 13. How many full packets can it make, and how many cards remain?

Day 005 | Unit 01 | Answer key

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1. 81

72 ÷ 9: write 8, subtract 72, remainder 0; bring down the next digit: 9 ÷ 9: write 1, subtract 9, remainder 0. Check: 9 × 81 = 729.

6.NS.B.2 | Constructed response

2. 63

25 ÷ 4: write 6, subtract 24, remainder 1; bring down the next digit: 12 ÷ 4: write 3, subtract 12, remainder 0. Check: 4 × 63 = 252.

6.NS.B.2 | Constructed response

3. 86

43 ÷ 5: write 8, subtract 40, remainder 3; bring down the next digit: 30 ÷ 5: write 6, subtract 30, remainder 0. Check: 5 × 86 = 430.

6.NS.B.2 | Constructed response

4. 121

22 ÷ 19: write 1, subtract 19, remainder 3; bring down the next digit: 39 ÷ 19: write 2, subtract 38, remainder 1; bring down the next digit: 19 ÷ 19: write 1, subtract 19, remainder 0. Check: 19 × 121 = 2299.

6.NS.B.2 | Constructed response

5. 194

27 ÷ 14: write 1, subtract 14, remainder 13; bring down the next digit: 131 ÷ 14: write 9, subtract 126, remainder 5; bring down the next digit: 56 ÷ 14: write 4, subtract 56, remainder 0. Check: 14 × 194 = 2716.

6.NS.B.2 | Constructed response

6. 156

59 ÷ 38: write 1, subtract 38, remainder 21; bring down the next digit: 212 ÷ 38: write 5, subtract 190, remainder 22; bring down the next digit: 228 ÷ 38: write 6, subtract 228, remainder 0. Check: 38 × 156 = 5928.

6.NS.B.2 | Constructed response

7. 319 R 2

67 ÷ 21: write 3, subtract 63, remainder 4; bring down the next digit: 40 ÷ 21: write 1, subtract 21, remainder 19; bring down the next digit: 191 ÷ 21: write 9, subtract 189, remainder 2. Check: 21 × 319 + 2 = 6701.

6.NS.B.2 | Constructed response

8. 157 R 13

23 ÷ 15: write 1, subtract 15, remainder 8; bring down the next digit: 86 ÷ 15: write 5, subtract 75, remainder 11; bring down the next digit: 118 ÷ 15: write 7, subtract 105, remainder 13. Check: 15 × 157 + 13 = 2368.

6.NS.B.2 | Model and constructed response

9. 305 R 27

113 ÷ 37: write 3, subtract 111, remainder 2; bring down the next digit: 21 ÷ 37: write 0, subtract 0, remainder 21; bring down the next digit: 212 ÷ 37: write 5, subtract 185, remainder 27. Check: 37 × 305 + 27 = 11312.

6.NS.B.2 | Model and constructed response

10. 370 full packets; 12 cards remain

4822 = 13 × 370 + 12.

6.NS.B.2 | Constructed response